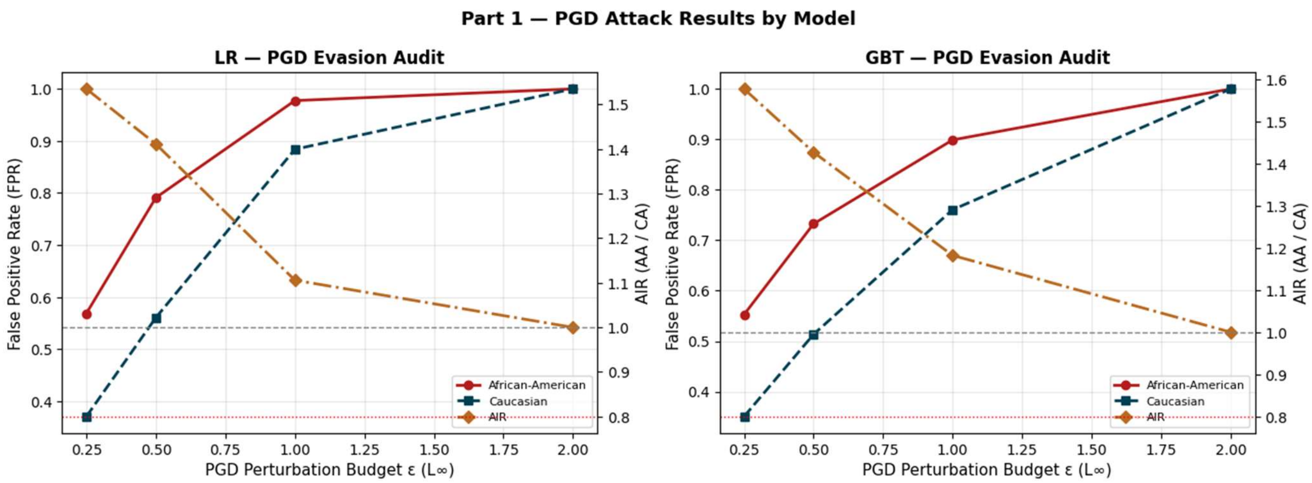
Individual Homework 05: Written Report

DNSC 6330: Responsible Machine Learning | Hesham Mushtaq, G23607459

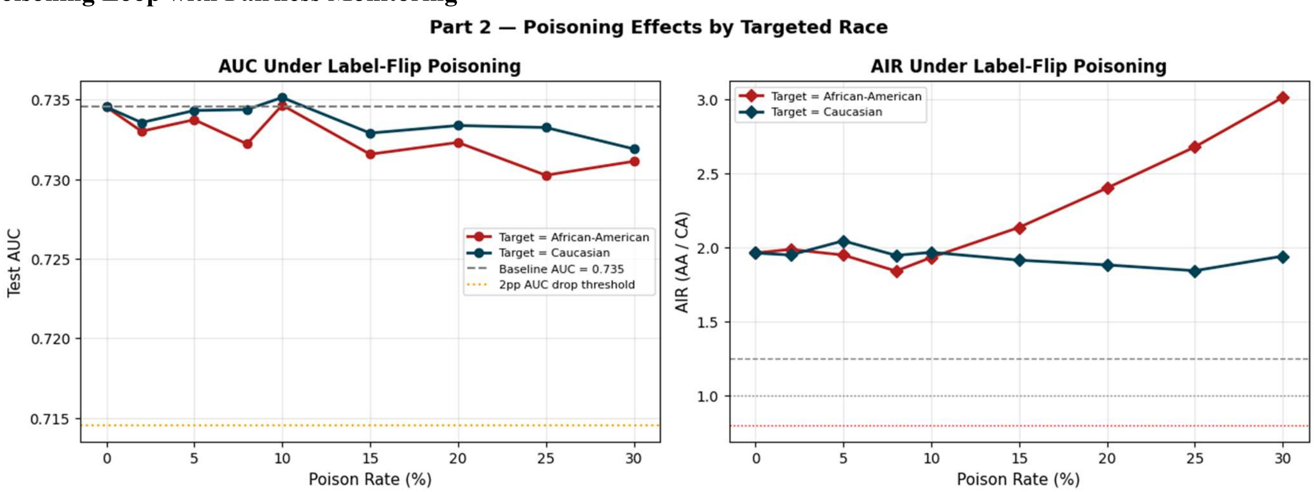
This report covers three attack pipelines on the COMPAS dataset: a PGD evasion audit, a label flip poisoning loop, and a shadow model membership inference. The clean baseline already showed unequal error rates by race. LR test AUC was 0.735 with FPR 0.281 (AA) and 0.143 (CA), AIR = 1.961. GBT test AUC was 0.718 with FPRs 0.317 and 0.178, AIR = 1.782. Every attack below acts on a model that was already imbalanced before any adversary touched it.

1. PGD Evasion Audit (LR and GBT)



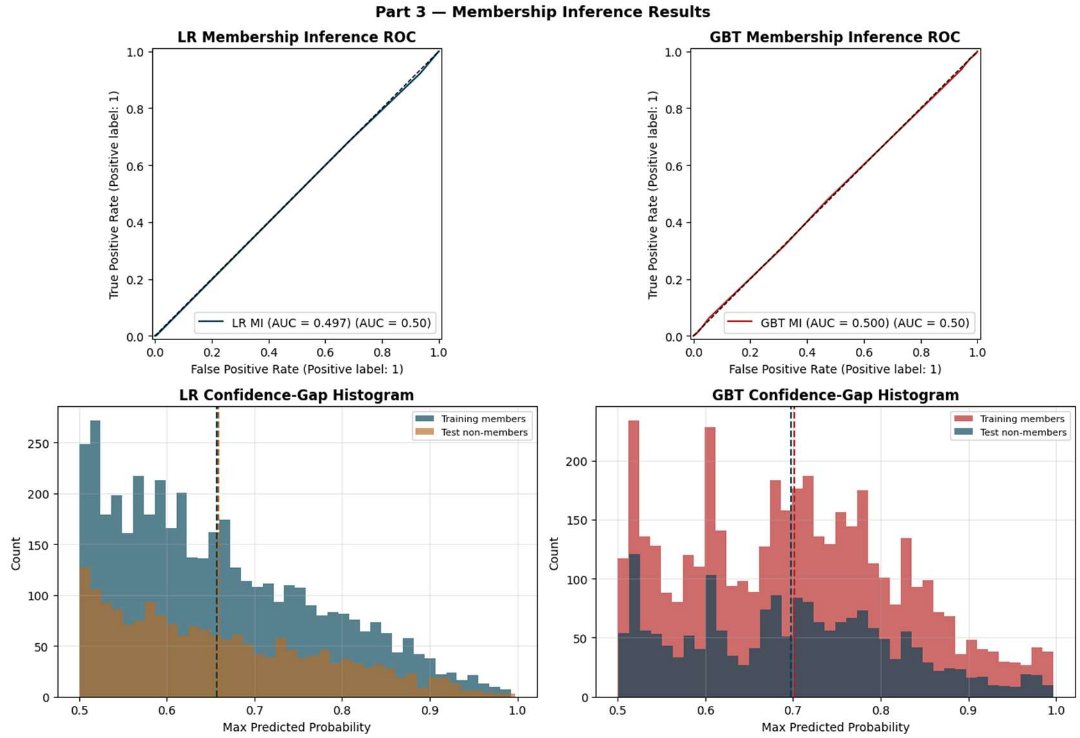
Both models break under PGD, but not at the same rate. LR FPRs (AA / CA) climb 0.569 / 0.370 at ε = 0.25, 0.791 / 0.560 at ε = 0.5, and 0.978 / 0.884 at ε = 1.0. GBT lags slightly: 0.553 / 0.351, 0.733 / 0.514, and 0.899 / 0.760 at the same budgets. By ε = 2.0 both models hit FPR = 1.000 across both groups total failure. The AIR curves move toward 1.0 in both panels, but that is not fairness improving; it is both groups being driven to the same bad outcome at the same time. Neither model has its AIR cross the 0.80 threshold at any tested ε, so the AIR crossing point lies outside the tested range. AIR alone would not flag this attack parity of failure looks identical to parity of success on this metric which is why subgroup FPR has to be monitored alongside it. Practically GBT looks like the safer pick, but neither is robust enough for high stakes deployment without extra protections.

2. Poisoning Loop with Fairness Monitoring

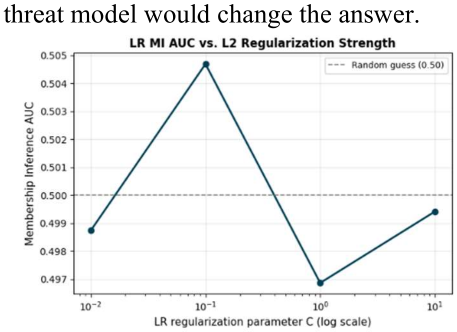


Test AUC barely moves across either target race variant, staying in roughly 0.730 to 0.735 a reviewer who only watches AUC would conclude the model is stable. AIR tells a different story. Targeting African American defendants pushes AIR from 1.961 at baseline to 3.010 at a 30% poison rate; the Caucasian targeted variant stays in roughly 1.84 to 2.04 across the range. On the stealth zone: the clean baseline already has AIR > 1.25, so the fairness band is violated before any poisoning happens. There is no clean threshold crossing point the entire tested range is effectively a stealth regime where AUC is flat but AIR is outside [0.80, 1.25] throughout. A PSI based drift monitor with the standard < 0.10 alert threshold would not catch either attack: every feature level PSI value comes out at 0.000 (max PSI = 0.000) since label flip poisoning leaves the feature distributions alone. AUC monitoring and PSI drift checks together would still miss this attack.

3. Membership Inference Depth



MI AUC is 0.497 for LR and 0.500 for GBT basically random guessing. ROC curves sit on the diagonal and the confidence gap histograms overlap heavily for both models, so the attacker is not separating training members from non members. The direction is consistent with the lecture point that overfitting helps MI: GBT has the larger generalization gap (+0.080) and the higher MI AUC, while LR has a small negative gap (0.008) and the lower MI AUC. The two point Pearson correlation is 1.000 by construction, which makes this a directional sanity check rather than a statistical test. An MI AUC near 0.50 also reports the average leakage but misses worst case record risk a single highly memorized outlier can leak even when the population level signal is flat and assumes a black box attacker without auxiliary data; a stronger threat model would change the answer.



Sweeping L2 over $C \in \{0.01, 0.1, 1.0, 10.0\}$ on LR keeps MI AUC essentially flat near 0.50, so stronger regularization (smaller C) buys no real privacy gain on this setup. The standard tradeoff smaller C reduces memorization but costs predictive performance is empty here because the privacy gain is zero across the full C range, so giving up utility buys nothing. The action this justifies is not "add regularization for privacy" but "don't pay a utility cost for a privacy gain that isn't there." DP training would only become justifiable if the model class were swapped for one that overfits more, or if the training set grew enough to expose memorization.

4. Conclusion

The highest risk finding is the PGD evasion result: a deployment time attack that flips outcomes without retraining anything. At $\varepsilon = 1.0$ LR hits FPR_AA = 0.978 / FPR_CA = 0.884 and GBT hits 0.899 / 0.760, with both collapsing to 1.000 by $\varepsilon = 2.0$. As a proactive mitigation, model selection should include adversarial stress testing rather than relying on clean AUC alone preferring GBT over LR drops FPR_AA by 7.9 pp and FPR_CA by 12.4 pp at $\varepsilon = 1.0$, which helps but does not solve the problem. As a reactive mitigation, monitoring should report subgroup FPR and AIR alongside AUC, since the poisoning result shows AUC stays at 0.731 at a 30% poison rate while AIR reaches 3.010 with PSI fixed at 0.000. The disparate impact catch is that switching models still leaves African American defendants with the higher FPR under attack, and fairness monitoring detects harm rather than preventing it so robustness and fairness controls must be used together.